UMJI FA22 VG101 Homework 7

Assigned: 15 Nov. 2022 Due: 22 Nov. 2022 23:59pm

Notes for Submission

Please submit to the JOJ for grading purpose (**follow the naming rule on JOJ**).

For backup purpose, you should also submit your homework to the Canvas. For canvas submission, please name the file using the format: `hw7-[Your Last Name]-[Your First Name]-[Your SJTU ID].c` For example, `hw7-Yang-Zijun-521370910109.c`

Homework Problem

Read the following program and complete the function `sortscores` to sort the students and their scores in ***descending*** order according to the scores. The input argument is a character string str, which is an array consists of names and scores of students, alternatively, separated by spaces. An example str with 3 students is "student1 100 student2 95 student3 97", which means that the scores of student1, student2, and student3 are 100, 95, and 97, respectively. We assume that:

1. Here is only one space between the names and numbers;
2. The first character and last character of the string are not spaces;
3. The students' names contain only letters or digits;
4. The scores are integers between 0-100;
5. The number of students ranges from 1 to 10;
6. The total number of characters in the input string is less than 255.

For example, if the user input is "student1 100 student2 95 student3 97", then the output should be "student1 100 student3 97 student2 95". You don't need to consider invalid inputs.

```c
#include <stdio.h>
#include <string.h>
// You can include libraries you need

// You can write other custom functions

// hw7 - finish this function
```

```
void sortscores(char str[]) {
}

// Do NOT change the following lines of code!!!
int main() {
    char str[256];
    scanf("%[^\n]%*c", str);  // ask for the user input
    sortscores(str);
    printf("%s\n", str);  // output
    return 0;
}
```

The following is an example run of the program:

